The exact value for the minimum detectable SNR depends on the *size* of the object; the larger the object, the easier it is to detect. To understand this, imagine smoothing the image in Fig. 25-8 with a 3×3 square filter kernel. This leaves the contrast the same, but reduces the noise by a factor of three (i.e., the square root of the number of pixels in the kernel). Since the SNR is tripled, lower contrast objects can be seen. To see fainter objects, the filter kernel can be made even larger. For example, a 5×5 kernel improves the SNR by a factor of $\sqrt{25} = 5$. This strategy can be continued until the filter kernel is equal to the size of the object being detected. This means the ability to detect an object is proportional to the *square-root* of its *area*. If an object's diameter is doubled, it can be detected in twice as much noise.

Visual processing in the brain behaves in much the same way, smoothing the viewed image with various size filter kernels in an attempt to recognize low contrast objects. The three profiles in Fig. 25-8 illustrate just how good humans are at detecting objects in noisy environments. Even though the objects can hardly be identified in the profiles, they are obvious in the image. To really appreciate the capabilities of the human visual system, try writing algorithms that operate in this low SNR environment. You'll be humbled by what your brain can do, but your code can't!

Random image noise comes in two common forms. The first type, shown in Fig. 25-9a, has a constant amplitude. In other words, dark and light regions in the image are equally noisy. In comparison, (b) illustrates noise that *increases with the signal level*, resulting in the bright areas being more noisy than the dark ones. Both sources of noise are present in most images, but one or the other is usually dominant. For example, it is common for the noise to decrease as the signal level is decreased, until a plateau of constant amplitude noise is reached.

A common source of constant amplitude noise is the video *preamplifier*. All analog electronic circuits produce noise. However, it does the most harm where the signal being amplified is at its smallest, right at the CCD or other imaging sensor. Preamplifier noise originates from the random motion of electrons in the transistors. This makes the noise level depend on how the electronics are designed, but not on the level of the signal being amplified. For example, a typical CCD camera will have an SNR of about 300 to 1000 (40 to 60 dB), defined as the full scale signal level divided by the standard deviation of the constant amplitude noise.

Noise that increases with the signal level results when the image has been represented by a small number of individual particles. For example, this might be the *x-rays* passing through a patient, the *light photons* entering a camera, or the *electrons* in the well of a CCD. The mathematics governing these variations are called **counting statistics** or **Poisson statistics**. Suppose that the face of a CCD is uniformly illuminated such that an average of 10,000 electrons are generated in each well. By sheer chance, some wells will have more electrons, while some will have less. To be more exact, the number of electrons will be normally distributed with a mean of 10,000, with some standard deviation that describes how much variation there is from